

Smart Water Level Monitoring for Industrial Tanks

A PROJECT REPORT SUBMITTED
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
MDC-Industry 4.0 -401

BACHELOR OF TECHNOLOGY

Minor Degree Course in

[Industry 4.0]

SEMESTER IV



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Program / Branch / Section	B.Tech [Electronics and communication]
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Submission Date	06 June 2026

SUBMITTED TO
Department of Electronics and Communication Engineering
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2026

CERTIFICATE

This is to certify that the project report entitled [Smart Water Level Monitoring for Industrial Tanks] submitted by [Vasu Jamwal (2024a3r019) and Ayush Sambyal (2024a3r018)] is a record of the work carried out for the Minor Degree Course [MDC-Industry 4.0 -401(Industry 4.0)] during Semester [IV]. The work has been completed under the guidance/supervision of the undersigned and is recommended for academic evaluation.

Date	06 June 2026	Signature of Faculty In-charge	_____
Place	MIET ,JAMMU	Name	Bhanu Pratap Singh Jamwal

STUDENT DECLARATION

We hereby declare that this report is based on the project/course work carried out by us for the Minor Degree Course. All external sources, datasets, images, code libraries, APIs, open-source tools, websites, research papers, and generative AI assistance, if used, have been properly acknowledged. We also declare that this work has not been submitted elsewhere for academic credit.

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ABSTRACT

Smart Water Tank Monitoring System

Water is a vital resource, yet a significant amount is wasted due to poor monitoring of water storage tanks. Manual checking of tank levels is time-consuming and often results in overflow, water wastage, and unnecessary electricity consumption. To address this issue, a Smart Water Tank Monitoring System has been developed using an Arduino Uno, ultrasonic sensor, LCD display, and buzzer.

The ultrasonic sensor continuously measures the distance between the sensor and the water surface. The Arduino processes this data and calculates the water level percentage in real time. The measured level is displayed on an LCD screen, enabling users to easily monitor tank status. When the water level reaches a predefined threshold, the buzzer generates an alert to prevent overflow and promote efficient water usage.

The system is cost-effective, easy to install, and suitable for households, schools, offices, and small industries. It contributes to sustainable water management by reducing wastage and encouraging responsible consumption of water resources.

KEYWORDS

Arduino Uno, Ultrasonic Sensor, Water Level Monitoring, Smart System, Sustainability, SDG 12.

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GENERAL FORMATTING AND WRITING RULES

Item	Recommended Rule
Target report length	Semester IV: 8-10 pages including preliminary pages. Semester VI: 10-12 pages including preliminary pages. Appendices may be additional if permitted.
Paper and margins	A4, portrait orientation, 1 inch margins. Add binding margin only if hard binding is required.
Font and spacing	Times New Roman, 11 or 12 pt body text, 1.15 or 1.5 line spacing, consistent paragraph spacing.
Headings	Chapter headings: 14 pt bold. Section headings: 12 pt bold. Use simple decimal numbering: 1, 1.1, 1.2.
Figures and tables	Use only relevant screenshots/diagrams. Explain every figure/table in the text. Avoid oversized screenshots.
Code snippets	Show only short and meaningful snippets in the main report. Full code should be in GitHub/appendix if required.
Citation style	Use IEEE numeric style [1], [2] unless the course faculty specifies another style.
Submission hygiene	Check grammar, plagiarism, citations, formatting, figure clarity, and page numbering before submission.

CHAPTER 1: PROJECT OVERVIEW

Water conservation is becoming increasingly important due to growing demand and limited availability. Many water tanks lack proper monitoring systems, leading to overflow and wastage.

1.1 Background and Motivation

Water is a critical resource required for daily activities in homes, institutions, and industries. However, many water storage tanks still rely on manual monitoring methods, which are inefficient and often lead to overflow, wastage, and unexpected shortages. With increasing water scarcity and rising demand, there is a growing need for smart solutions that enable efficient water management.

This project belongs to the domain of **IoT-based Technologies and Embedded Systems**, where sensors and microcontrollers are used to monitor and automate real-world processes. The primary users include households, educational institutions, commercial buildings, and small industries that depend on water storage tanks.

The motivation behind this project is to develop a simple, affordable, and effective system that provides real-time water level monitoring and alerts users before overflow occurs. The topic is well-suited for the Minor Degree Course as it integrates hardware interfacing, sensor technology, embedded programming, and sustainability concepts while addressing a practical real-world problem.

1.2 Problem Statement

Manual monitoring of water tanks is inefficient and often causes overflow, water wastage, and unexpected shortages.

1.3 Objectives

- Monitor water levels in real time.
- Display water level percentage on LCD.
- Provide overflow alerts using a buzzer.
- Reduce water wastage.
- Promote sustainable water management.

1.4 Scope and Expected Learning Outcomes

The system can be used in homes, schools, offices, and institutions for effective water tank monitoring.

Expected Learning Outcomes

- Understand the fundamentals of embedded systems and sensor interfacing.
- Gain practical experience with Arduino programming and hardware integration.
- Learn real-time data acquisition and processing techniques.
- Develop skills in circuit design, testing, and troubleshooting.
- Understand the application of technology in sustainable water management.
- Improve problem-solving, project development, and system implementation skills.

CHAPTER 2: BACKGROUND AND TOOLS USED

Traditional water tank monitoring methods require manual inspection and are prone to errors. Smart monitoring systems provide automated and accurate measurements.

2.1 Essential Background

The Smart Water Tank Monitoring System is based on embedded systems and sensor technology. An **ultrasonic sensor (HC-SR04)** is used to measure the distance between the sensor and the water surface by transmitting ultrasonic waves and calculating the time taken for the echo to return. This distance is then used to determine the water level in the tank.

The system uses an **Arduino Uno microcontroller**, which processes the sensor data and converts it into a water level percentage. The calculated level is displayed on a **16×2 LCD display**, allowing users to monitor the tank status in real time. A **buzzer** is incorporated to provide an audible alert when the water level reaches a predefined threshold, helping to prevent overflow and water wastage.

Embedded monitoring systems are widely used in smart infrastructure and IoT applications because they offer automation, real-time monitoring, and improved resource management. By integrating sensors, controllers, and display units, such systems enhance efficiency while reducing human intervention.

2.2 Tools, Technologies and Dataset / Input Used

Category	Details
Programming Language	Embedded C / Arduino C
Framework / Library	Wire.h, LiquidCrystal_I2C, NewPing
Database / Storage	Not Applicable (Real-time monitoring system)
Dataset / Input Source	Real-time distance measurements from HC-SR04 Ultrasonic Sensor mounted on the water tank
Tools	Arduino IDE / ArduinoDroid, Arduino Uno, Ultrasonic Sensor (HC-SR04), LCD Display (16×2 I2C), Breadboard, Jumper Wires, Buzzer

2.3 Mini Gap / Need Analysis

Existing Approach / Source	Main Feature	Limitation / Need	Project Response
Manual Water Tank Monitoring	Users manually check water level in tanks	Time-consuming, inaccurate, and may cause overflow or water shortage	Provides automatic real-time water level monitoring and display
Commercial Water Monitoring Systems	Automated monitoring and alerts	Expensive, complex, and less affordable for small households	Offers a low-cost, simple, and easy-to-install monitoring solution using Arduino and ultrasonic sensor

CHAPTER 3: DESIGN AND IMPLEMENTATION

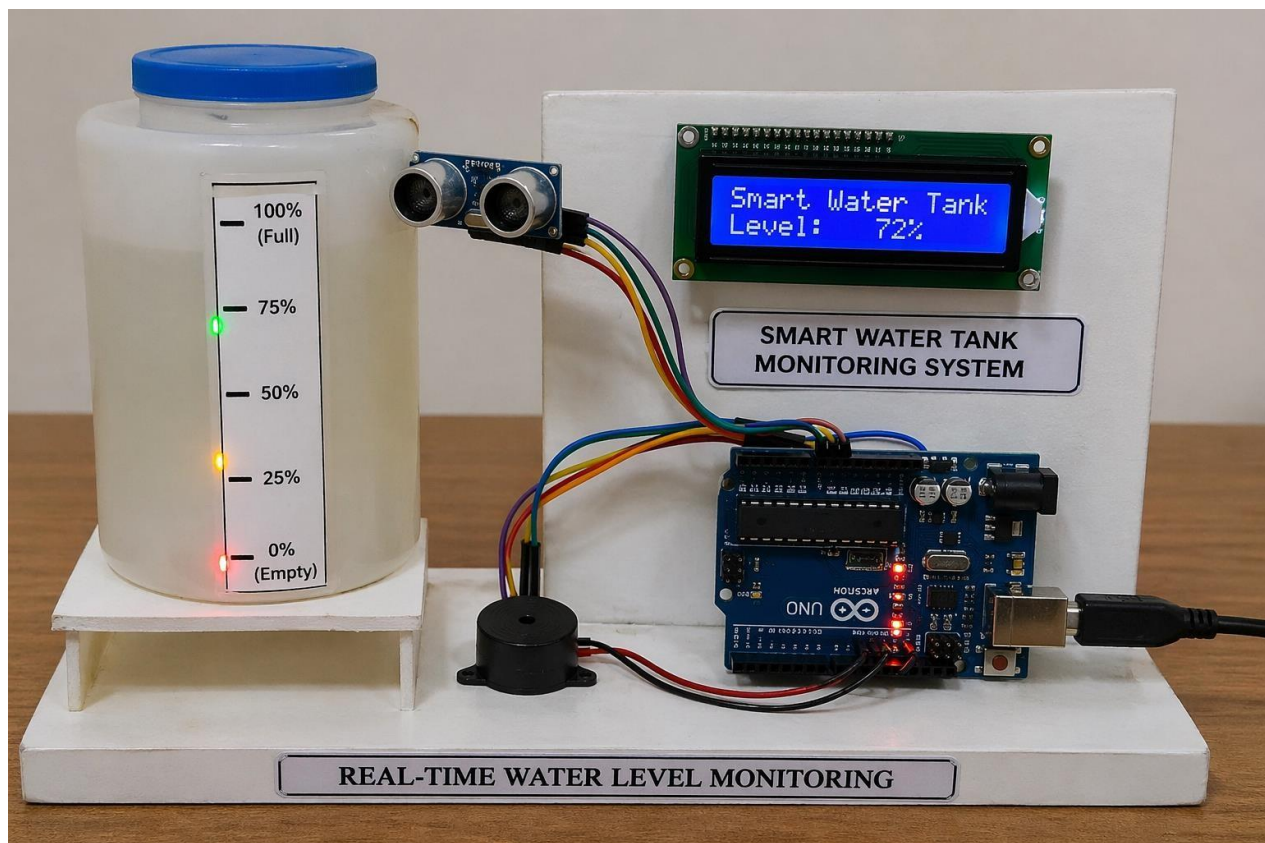
3.1 Functional Requirements

Req. ID	Requirement	Priority
FR-01	The system shall measure the water level in the tank using an ultrasonic sensor.	High
FR-02	The system shall display the water level percentage on the LCD screen.	High
FR-03	The system shall provide an alert through a buzzer when the water level reaches a predefined threshold.	High
FR-04	The system shall continuously monitor and update water level readings in real time	Medium

3.2 Methodology / Workflow

The system uses an ultrasonic sensor mounted at the top of the water tank to measure the distance between the sensor and the water surface. The measured distance is sent to the Arduino Uno, which processes the data and converts it into a water level percentage. The calculated level is displayed on the LCD screen for real-time monitoring. When the water level exceeds the predefined threshold, the buzzer generates an alert to prevent overflow. The system continuously repeats this process, ensuring accurate and updated readings.

3.3 System Architecture / Workflow Diagram



Ultrasonic Sensor → Arduino Uno → Data Processing → LCD Display & Buzzer Alert

3.4 Module Description

Module	Responsibility	Input / Output
Sensor Module	Measures distance from water surface	Input: Water level, Output: Distance value

Processing Module (Arduino Uno)	Calculates water level percentage and controls alerts	Input: Sensor data, Output: Water level percentage
Display & Alert Module	Displays level and provides warning alerts	Input: Processed data, Output: LCD display and buzzer alert

3.5 Key Implementation Details

- ☐ Arduino Uno is used as the main controller.
- ☐ HC-SR04 ultrasonic sensor measures water level using sound wave reflection.
- ☐ Water level is calculated based on tank height and measured distance.
- ☐ LCD (16×2 I2C) displays the current water level percentage.
- ☐ Buzzer generates an alert when the tank reaches the specified threshold level.
- ☐ The system operates in real time and continuously updates measurements.
- ☐ The prototype is implemented using Arduino IDE/ArduinoDroid and tested on a model water tank.

3.6 Important Code Snippet / Pseudocode

The following pseudocode represents the working logic of the system:

Start

Read distance from ultrasonic sensor

Calculate water level percentage

Display water level on LCD

If water level $\geq$ threshold

 Activate buzzer

Else

 Keep buzzer OFF

Repeat continuously

End

The algorithm continuously monitors the water level, updates the display, and generates alerts when the tank is close to overflowing.

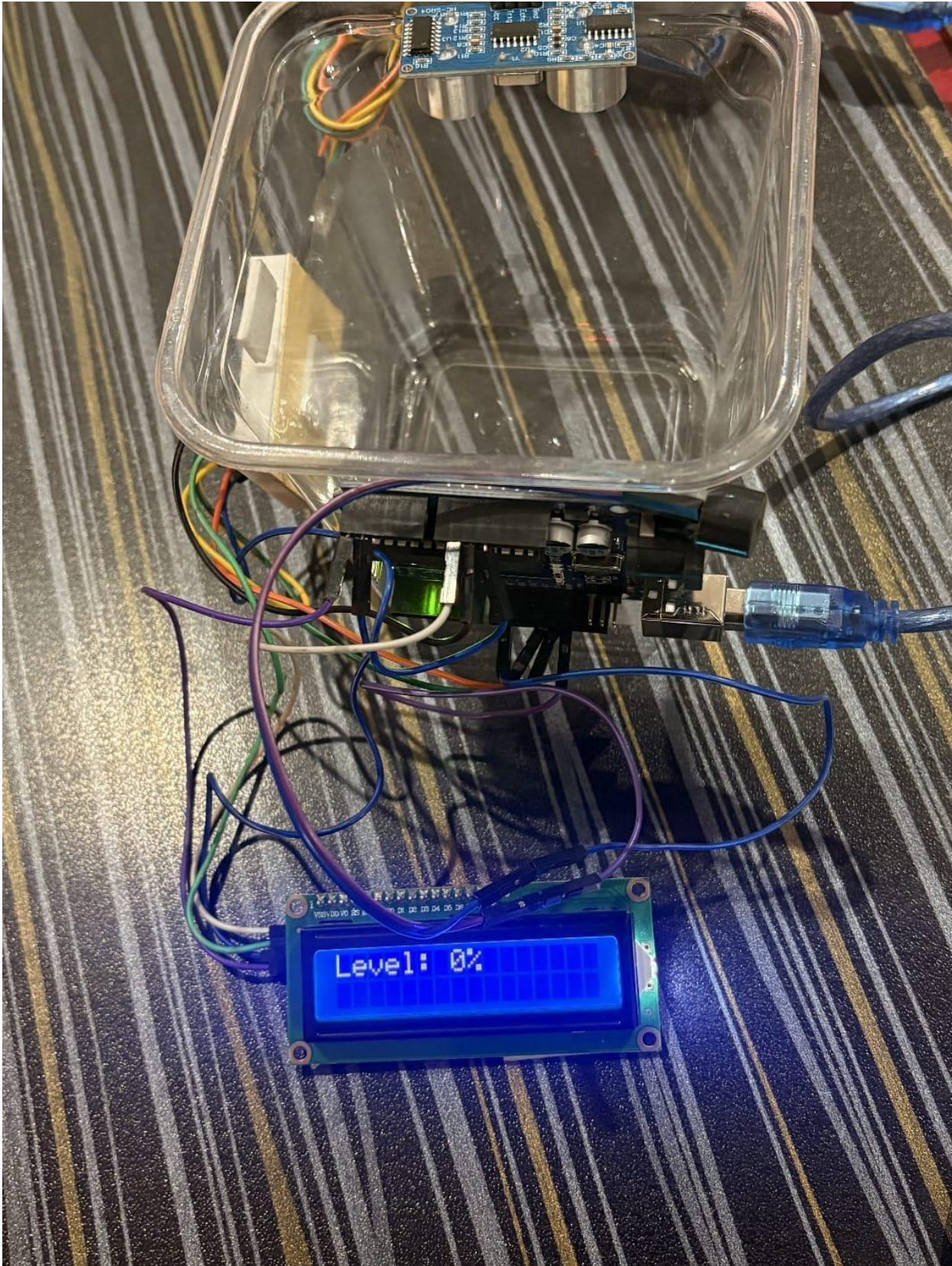
CHAPTER 4: RESULTS, TESTING AND DISCUSSION

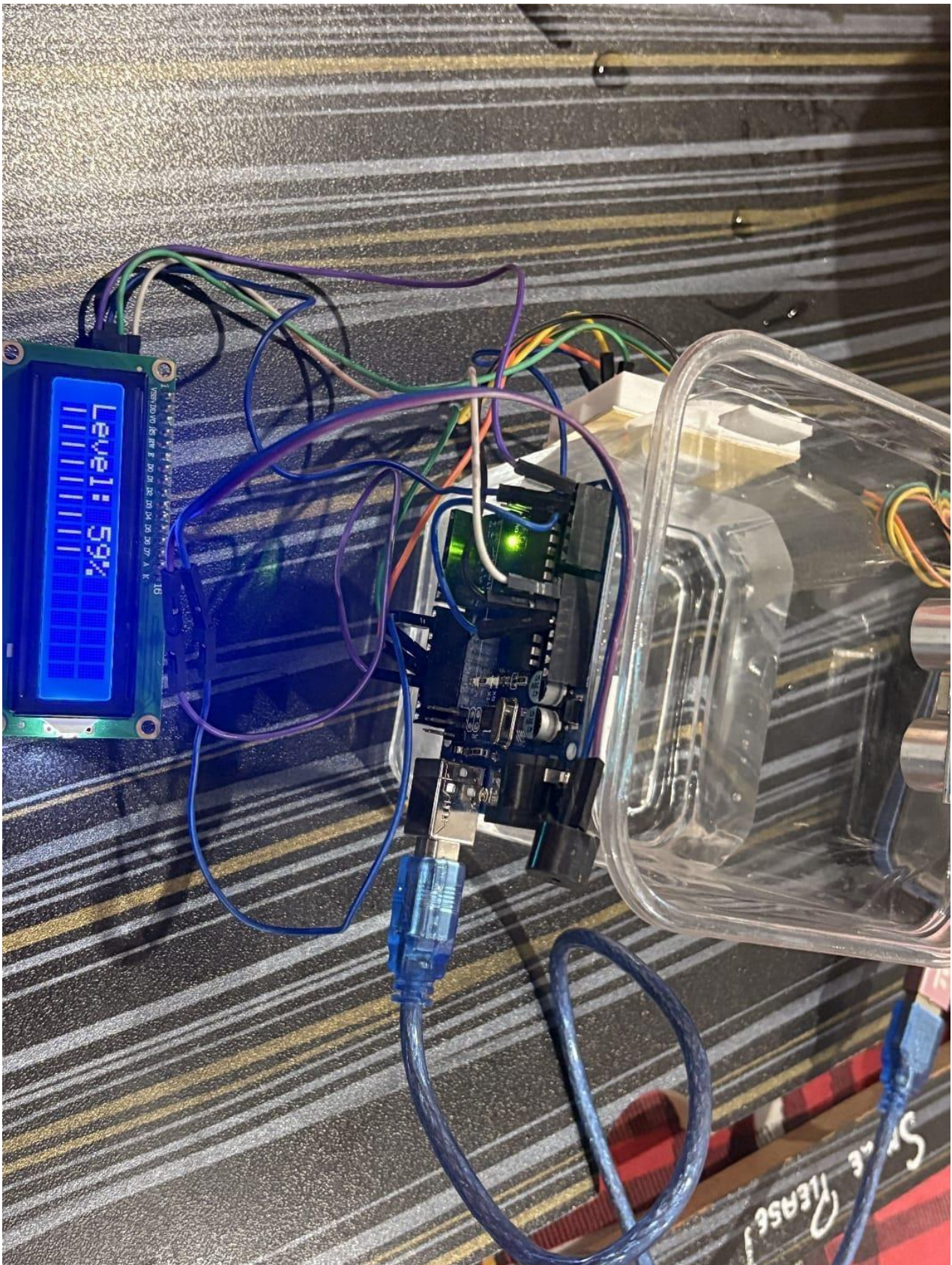
4.1 Test Cases

Test ID	Scenario	Input	Expected Output	Result
TC-01	Tank Empty Condition	Empty tank (0–10% water level)	LCD displays low percentage, buzzer OFF	Pass
TC-02	Medium Water Level	Water level around 50%	LCD displays approximately 50% level	Pass
TC-03	Tank Nearly Full	Water level above 80%	LCD displays high percentage and buzzer sounds for 3 seconds	Pass
TC-04	Continuous Monitoring	Changing water level inside tank	LCD updates water level in real-time	Pass

4.2 Output Screens / Experimental Results

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Experimental Results

Actual Water Level Displayed Water Level

Empty Tank 0–10%

Half Filled Tank 45–55%

Actual Water Level Displayed Water Level

75% Filled Tank 70–80

Full Tank 90–100%

The ultrasonic sensor successfully measured the distance between the sensor and water surface. The Arduino Uno processed the readings and calculated the corresponding water level percentage. The LCD displayed the measured level in real time, while the buzzer generated an alert whenever the tank level exceeded the predefined threshold.

4.3 Discussion and Observation

The developed Smart Water Tank Monitoring System successfully achieved its objective of monitoring water levels in real time. The ultrasonic sensor provided reliable distance measurements, which were converted into water level percentages using Arduino programming. The LCD display continuously updated the water level information, allowing users to easily monitor tank status.

The buzzer alert mechanism effectively notified users when the tank reached a high water level, helping prevent overflow and water wastage. During testing, minor fluctuations in readings were observed due to sensor noise and reflections from the tank surface. This issue was reduced by averaging multiple sensor readings in software.

Overall, the system demonstrated good accuracy, low cost, ease of implementation, and suitability for household and small-scale water management applications. The project also provided practical learning in embedded systems, sensor interfacing, Arduino programming, and real-time monitoring techniques.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Smart Water Tank Monitoring System was successfully designed, implemented, and tested using an Arduino Uno, HC-SR04 ultrasonic sensor, LCD display, and buzzer. The system measures the water level inside a tank in real time and displays the corresponding percentage on the LCD screen. An alert is generated through the buzzer when the water level reaches a predefined threshold, helping to prevent overflow and water wastage.

The project achieved the objectives defined in Chapter 1 by providing a low-cost, user-friendly, and automated solution for water level monitoring. Through this project, practical knowledge was gained in sensor interfacing, embedded systems, Arduino programming, real-time data processing, and hardware-software integration. The developed prototype demonstrates the feasibility of smart water management systems for domestic and institutional applications.

5.2 Limitations

- The prototype was tested only on a small-scale model tank and not in a real-world large water storage environment.
- Ultrasonic sensor readings may fluctuate slightly due to environmental factors, surface reflections, or sensor positioning.
- The current system provides local monitoring only and does not support remote monitoring or cloud-based data storage.
- The project uses basic alert functionality and does not automatically control water pumps.

5.3 Future Scope

- Integration with IoT platforms such as Wi-Fi or GSM modules for remote monitoring through mobile applications.
- Automatic control of water pumps based on tank water level to achieve complete automation.
- Cloud-based storage and analysis of water consumption data for better resource management.
- Development of a dedicated mobile dashboard with notifications, analytics, and historical records.
- Use of advanced sensors and filtering algorithms to improve measurement accuracy and reliability.

5.4 Ethical, Security and Social Considerations

The system is designed to promote responsible water usage and reduce water wastage, contributing positively to environmental sustainability. Since the current prototype does not collect personal information, privacy concerns are minimal. However, if future IoT connectivity is added, appropriate security measures such as authentication, encrypted communication, and secure data storage should be implemented to protect user data.

The system should be used responsibly and maintained regularly to ensure accurate measurements and safe operation. The low-cost design improves accessibility and makes smart water monitoring technology available to a wider range of users, including households, schools, and small institutions.

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FINAL SUBMISSION CHECKLIST

Check	Status
Title is specific and matches actual work.	Done
Cover page contains correct minor course name, course code, semester, branch/section, and faculty details.	Done
Certificate and declaration are complete and signed by all group members.	Done
Abstract states problem, method, output/result, and relevance.	Done
Objectives are measurable and reflected in results/conclusion.	Done
Background/tools are concise and properly cited.	Done
Architecture/workflow diagram is readable and explained.	Done
Implementation section explains important decisions, not only code.	Done
Test cases/results are presented and discussed.	Done
Figures and tables are numbered and referred to in the text.	Done
References follow a consistent style.	Done
External code, datasets, APIs and AI tools are acknowledged.	Done
Grammar, formatting, alignment and page numbering have been checked.	Done

BASIS OF THIS MODIFIED TEMPLATE

This project report has been prepared using the modified Minor Degree Course (MDC) project report template provided by the institution. The template is designed to present the project work in a structured and concise manner while covering all essential academic components such as the abstract, objectives, methodology, implementation, testing, results, conclusion, and references. The format ensures proper documentation of the project and helps maintain uniformity across all student submissions. This report follows the prescribed guidelines and presents the development and evaluation of the IoT-Based Water Tank Monitoring System in a clear and organized manner.